Design and simulation of XOR, TSPC, and MUX based Hybrid phase detectors in 45nm CMOS Technology

Pavana P
Electronics & Communications
BNM Institute of Technology
Bengaluru
[1]pavana.gooli@gmail.com

Divya P
Electronics & Communications
BNM Institute of Technology
Bengaluru
[2] p.divyadivya2006@gmail.com

Poorvika M R
Electronics & Communications
BNM Institute of Technology
Bengaluru
poorvikamr305@gmail.com

Simtha Gayathri
Electronics & Communications
BNM Institute of Technology
Bengaluru
@gmail.com

***Abstract*-Phase-Locked Loops (PLLs) are fundamental to signal synchronization and frequency synthesis in modern digital and communication systems. At the heart of the PLL, the phase detector (PD) measures the phase difference between two signals. While simple XOR-based PDs are effective for phase differences between 0° and 180°, their accuracy diminishes significantly beyond this range. To overcome this limitation, this paper proposes a hybrid phase detector architecture combining XOR and True Single Phase Clocked (TSPC) logic. This hybrid approach leverages both detection methods to achieve comprehensive, full-range (0°–360°) phase detection, enhancing both accuracy and reliability. The design benefits from the rapid locking response of latch-based PDs and the low-power, low-jitters characteristics of TSPC logic once the phase is locked. This work introduces an innovative multiplexer (MUX) based design that provides a balanced, scalable, and modern PLL implementation capable of accurate phase tracking across its entire operational range.**

Keywords: **PLL, PD, XOR, TSPC, MUX, Hybrid Design, 45nm CMOS.**

I. INTRODUCTION

Phase-Locked Loops (PLLs) are essential components in contemporary electronic and communication systems, providing precise synchronization of an output signal to a reference clock. They are foundational to frequency synthesizers and see wide application in clock recovery circuits, digital communication links, and signal processing. A typical PLL comprises a phase detector (PD), a loop filter, a voltage-controlled oscillator (VCO), and often, a frequency divider.

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Among these elements, the PD is the most crucial, as it dictates the PLL's ability to align the output phase with the reference signal and the time it takes to achieve this lock. For low-frequency digital systems, traditional PDs like the exclusive-OR (XOR) based detector are often sufficient due to their simplicity. The XOR detector converts the phase deviation between two periodic signals into a duty-cycle-modulated signal.

However, XOR-based detectors suffer from a limited linear operational range (0°–180°) and "dead zones" when there is no phase difference. These non-linear characteristics are detrimental in high-frequency applications, as they can slow the lock time and increase jitter.

To address these limitations, True Single-Phase Clock (TSPC) circuits offer a more efficient, energy-saving solution. A TSPC detector uses dynamic CMOS logic and operates with a single clock phase, which eliminates clock skew and reduces the required transistor count. This configuration is highly effective for phase detection in advanced deep submicron processes, such as 45nm CMOS. While TSPC PDs perform well at high speeds by lowering static power dissipation, they can be more susceptible to process variations and noise.

An additional methodology is the MUX-based (multiplexer-controlled) phase detector. This approach uses logic-level selection to dynamically switch between different phase-sensing techniques, enabling operation across a broader phase range. A hybrid PD can be created by merging the XOR detector's coarse detection capabilities with the TSPC circuit's fine-resolution control, using a MUX to switch between them. This arrangement can provide a broad linear range (0°–360°), improved phase noise, and dynamic power scalability.

The XOR phase detector is simply exclusive OR gate. We know the simple operation of exclusive OR gate, it compare phase difference between both signal and produce output pulses basis on the input varies. It provide dc level proportional to the phase difference between the inputs. While the XOR circuit produces error pulses on both rising and falling edges. The operation of phase detectors is similar to that of differential amplifiers in that both sense the difference between the inputs, generating a proportional output. It doesn't contain any phase difference information.

The phase difference between the dclock and data is given by

$$\Delta\oint = \oint data - \oint dclock = \frac{\Delta t}{Tdclock}.2\pi(radians)$$

or, in terms of output clock frequency

$$\Delta\oint = \frac{\Delta t}{2\,Tclock}.2\pi$$

This study details the design and simulation of a hybrid XOR-TSPC-MUX phase detector in 45nm CMOS technology. The control logic, based on a D flip-flop, dynamically switches between the two PD types. The fast XOR PD operates during the initial acquisition phase, and the system switches to the TSPC circuit for steady operation once in a locked state.

II. RELATED WORK

The evolution of integrated-circuit technology has continually expanded the capabilities of phase detectors, driven by the PLL demand for higher frequencies, lower jitter, and reduced power consumption. Trends in existing literature provide a basis for the hybrid XOR-TSPC-MUX design optimized for 45nm CMOS.

Khan et al. [1] and Banerjee et al. [2] were early contributors in 3D VLSI design, exploring efficient layer assignment and partitioning for complex multi-layer circuits. While not focused directly on phase detection, their work established a foundation for mitigating interconnect delay and power dissipation in deep-submicron nodes. This is critical for implementing high-speed PLL components, like PDs, in advanced nodes such as 45nm.

The need for ultra-low-power, high-reliability circuits was highlighted in studies on wireless sensor networks (WSNs) by Moussa et al. [3] and IoT-based monitoring systems by Sungheetha and Rajesh [4]. PLLs are central to achieving the precise clock synchronization these applications require. These studies underscored the need for energy-efficient PDs that can operate stably across a broad phase range.

Majeed and Kailath [5] made a more direct contribution by proposing spike-free phase-frequency detectors (PFDs) operating up to 900 MHz. Their design used multiplexing and reset logic to remove dead zones, thereby improving lock precision. Trivedi and Yadav [6] advanced this concept by creating a TSPC-based sequential PFD in 90nm CMOS that operated at 1 GHz using only 471 µW of power. These investigations inspired the TSPC component of the current hybrid detector, demonstrating that single-phase clocked dynamic logic offers superior speed-power trade-offs.

Wang et al. [7] introduced a 4:1 MUX-based frequency discriminator, which achieved a 30 dB reduction in phase noise by substituting digital multiplexers for analog multipliers. Their findings, which showed that MUX architectures can greatly increase the linear detection range, are clearly reflected in the MUX-controlled hybrid PD proposed in this work.

Phillimon and Devi [8] provided a thorough comparison of phase-measurement methods on FPGA platforms. They noted that while XOR-based detectors offer picosecond-level precision, their performance degrades beyond 180°. This limitation strongly encourages the integration of other techniques, like TSPC, for full-range operation. Fu et al. [9] demonstrated a nonlinear PFD using TSPC D-flip-flop structures, highlighting that combining nonlinear phase discrimination with optimized analog circuitry can result in higher lock stability and power efficiency.

Finally, Lukichev [10] used an XOR-gate-based PD to synchronize a generator with the electrical grid. This study demonstrated that while XOR gates are effective for fast phase comparison up to 180°, they are insufficient for full 360° coverage, reinforcing the need for hybrid designs.

III. Methodology And Implementation

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The goal of this work is to design and simulate a hybrid phase detector (PD) that merges the advantages of XOR, TSPC, and MUX architectures. This approach aims to achieve a broad detection range, low power consumption, and high accuracy, overcoming the limitations of using these components in isolation.

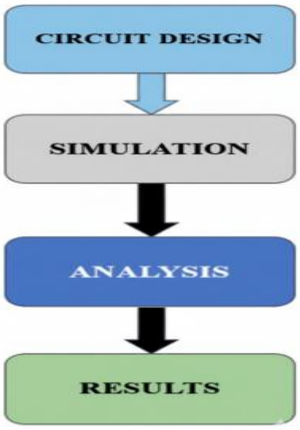

Fig 3.1: Design and Simulation Process Flowchart

The design and implementation process follows four primary phases, as shown in **Fig 3.1**:

1. **Circuit Design:** The individual blocks (XOR, TSPC, MUX, and control logic) are designed at the transistor level using 45nm CMOS technology within the Cadence Virtuoso platform.
2. **Simulation:** The completed design is verified using the Cadence Spectre environment. This involves transient analysis to test signal behavior and phase detection performance.
3. **Analysis:** The simulated outputs are evaluated to examine key parameters such as the phase detection range, power consumption, jitter, and lock time.
4. **Results:** The performance of the hybrid design is compared against traditional techniques to validate its effectiveness.

A conventional Phase-Locked Loop (PLL) is composed of a phase detector, a loop filter, and a voltage-controlled oscillator (VCO). The PD is responsible for identifying the phase difference between the reference signal and the VCO's feedback signal. While simple XOR-based PDs are often used for their speed and simplicity, they are hampered by a narrow operational range, typically limited to 0°–180°, which prevents full-phase detection.

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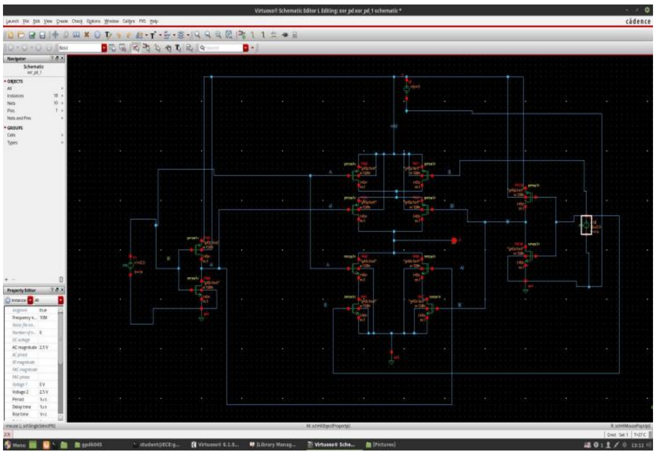

Fig 3.2: XOR Phase Detector

This schematic shows the testbench used to simulate the TSPC block. The block is powered by a 1.2V DC source and driven by two pulse voltage sources (V1 and V2) with a 10-picosecond rise/fall time. This setup is used for transient simulation to analyze its high-speed timing and operation.

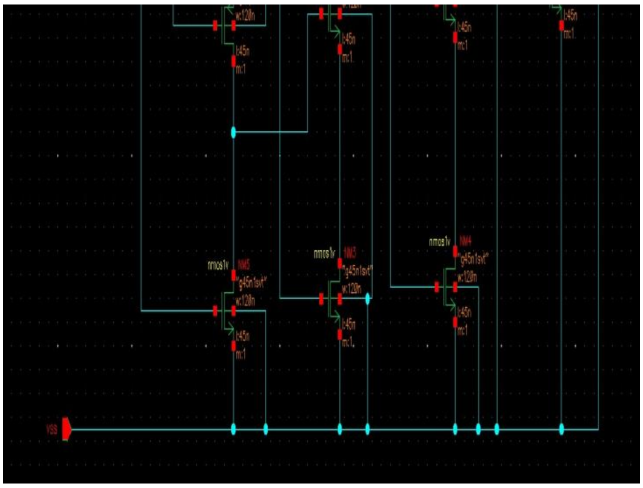

Fig 3.3: TSPC Phase Detector

The picture shows a section of an integrated circuit schematic with several NMOS (n-type Metal-Oxide-Semiconductor) transistors linked together in a typical arrangement, most likely a digital logic gate/buffer array or a current mirror array. Specific design specifications, such as a channel of width 120nm and channel length 45nm, define each transistor, which is referred to by names such as NMOS1V, NM5, and NM4. These transistors' sources are all connected to the VSS node, which stands for the circuit's ground or negative power supply rail, while their drains are connected to higher-level circuitry (not completely displayed). According to the connection, these transistors are made to function as switches or as current sinks in a larger CMOS or comparable low-voltage digital/analog system.

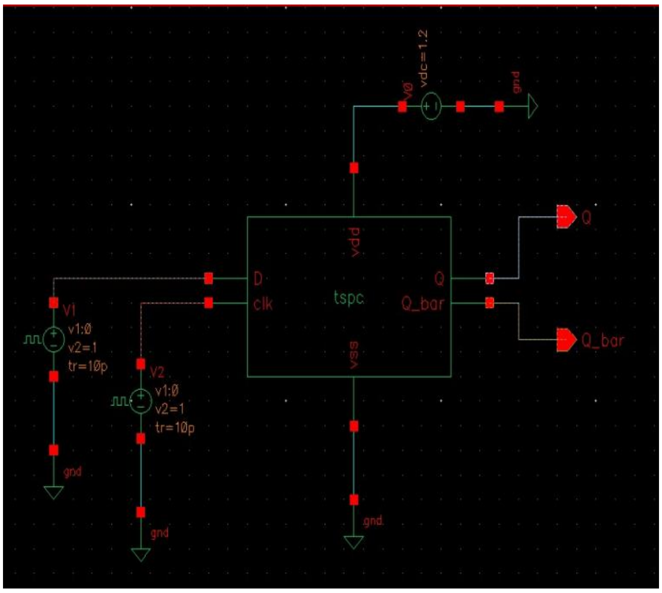

Fig 3.4 :TSPC Phase Detector Symbol

The picture shows a test bench schematic for modeling a digital logic component, more precisely a sequential circuit block with the name "tspc" (perhaps a register or flip-flop for a True Single-Phase Clock). The block has Q and its counterpart Q bar as outputs and Data(D) and Clock(clk) as inputs. The VDD terminal is connected to a DC voltage source ($\text{vdc}=1.2$ V) that is referenced to ground at the VSS terminal.Two pulse voltage sources (V1 and V2), each with a 1-volt amplitude and a 10-picosecond ($\text{tr}=10\text{p}$) rise/fall time, drive the inputs D and clk, indicating that the circuit is configured for a transient simulation to examine its timing and operation at high speeds. To observe the resulting Q and Q bar waveforms, the outputs are linked to probes or measurement elements (shown by the triangle symbols).

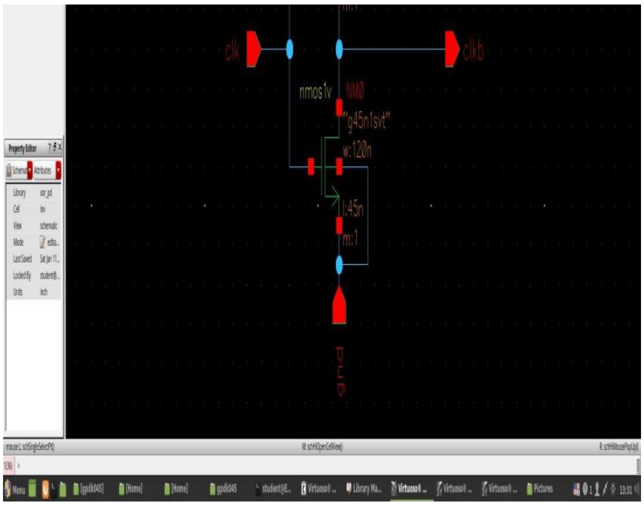

Fig 3.5: Inverter

The picture shows a simplified schematic fragment of a CMOS integrated circuit architecture with many external nodes coupled to a single NMOS transistor. One essential part is the transistor, which has two key technological characteristics: a channel of width 120nm with channel length of 45nm. The transistor's gate is connected to a node called clk, indicating that the clock signal controls its operation like a switch. The drain is connected to a node called clkb (clock-bar or inverted clock), and the source is connected to gnd. This arrangement suggests that it is probably a part of a pass-gate, dynamic logic gate, or inverter, where the NMOS pulls a node to ground when the clock is high.

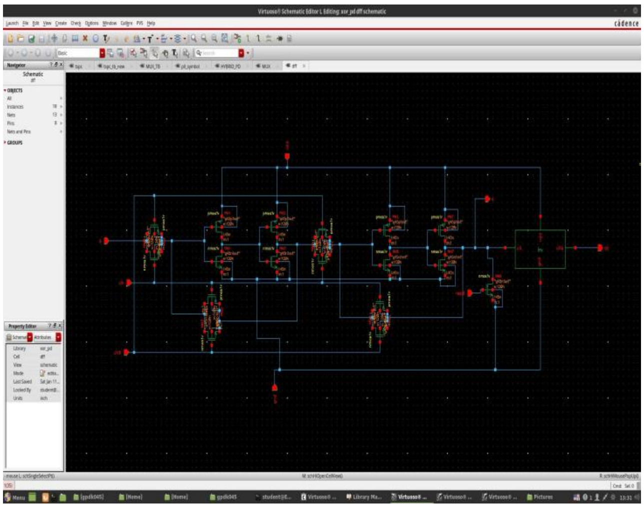

Fig 3.6: D Flip Flop

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It shows the transistor-level schematic of a digital logic circuit created with the Cadence Virtuoso Schematic Editor. Based on the file name "xor_pd.schematic," it is most likely an XOR or XNOR gate. Because the stages are formed by pairs of PMOS (p-type, top half) and NMOS (n-type, bottom half) transistors, the circuit is implemented in CMOS (Complementary Metal-Oxide-Semiconductor) logic. To accomplish the intended XOR/XNOR function, the structure displays a cascading arrangement of logic blocks, which could be a combination of regular CMOS gates or a cascaded pass-transistor logic. The numerous transistors and intricate routing point to a highly optimized or high-performance design that includes components like pre-drivers or buffers (inferred by the stages) to drive the output, which is seen on the right.

IV. RESULTS AND ANALYSIS

The transient response of the hybrid phase detector simulated in Cadence Virtuoso is represented by this waveform. The reference signal, the MUX select line, the inputs and outputs of the XOR and TSPC phase detectors, and the final hybrid PD output (`pd_out`) are among the signals displayed. The XOR detector's input is shown by the red waveform, and its matching output pulses, which are active when the phase difference is modest (0°–180°), are shown by the green waveform. The cyan waveform is the TSPC block's output, which is active during bigger phase differences (180°–360°), whereas the magenta signal is the input to the TSPC detector. The reference input signal that sustains synchronization and guarantees phase comparison during the process is indicated by the yellow waveform.

Fig 4.1:Waveform for phase between 0-180 degrees

When the phase difference is less than 180°, the MUX selects the XOR output, generating pulses that correspond to the behavior of the XOR PD; when the phase exceeds 180°, the TSPC output is selected, guaranteeing continuous phase detection without dead zones; this illustrates the adaptive switching mechanism of the hybrid PD, which combines the high-speed response of XOR logic with the low-power, wide-range detection of TSPC logic achieving full 0°–360° operation efficiently in 45 nm CMOS technology.

Fig 4.2: Waveform for phase between 180-360 degrees

It presents the waveform when the phase difference between the two input signals falls within the range of 180° to 360°. In this condition, the D flip-flop detects the phase relation and outputs a logic high (‘1’), indicating that the signals are out of phase beyond 180°. This high output from the D flip-flop serves as the select line for the multiplexer (MUX). With the select line set to ‘1’, the MUX switches to its alternate input, which is connected to the TSPC (True Single-Phase Clocked) logic output. Consequently, the system output now follows the behavior of the TSPC logic, allowing accurate phase detection and response in the 180°–360° range. This switching mechanism ensures that the system dynamically adapts to different phase ranges, enabling robust and precise interference analysis.

V. Conclusion

This research successfully demonstrated the design and simulation of a MUX-based hybrid Phase Detector (PD) for modern Phase-Locked Loop (PLL) systems. The design effectively combines the advantages of XOR-based and TSPC-based PDs by dynamically alternating between them based on the system's locking state.

This hybrid strategy harnesses the fast locking speed of the XOR PD for rapid acquisition and the low-power, low-jitter performance of the TSPC PD for steady-state operation. Simulation results confirm that the MUX-based hybrid PD

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offers superior performance in locking precision, jitter reduction, and power consumption compared to standalone XOR or TSPC designs. The adaptive mechanism of this hybrid detector makes it a well-rounded and efficient solution for applications where speed and power economy are critical, such as high-speed communication, IoT, automotive systems, and AI hardware.

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